TEAM RAZORSEDGE

Autobelay

long premium, short leash

An autonomous options agent on Alpaca's MCP server. An auto belay catches a climber with nobody holding the other end. Here the brake is code.

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The thesis, in one sentence

Buy defined-risk, short-dated options premium when an open model can **name a reason** — and let deterministic code size it, stop it, and close it.

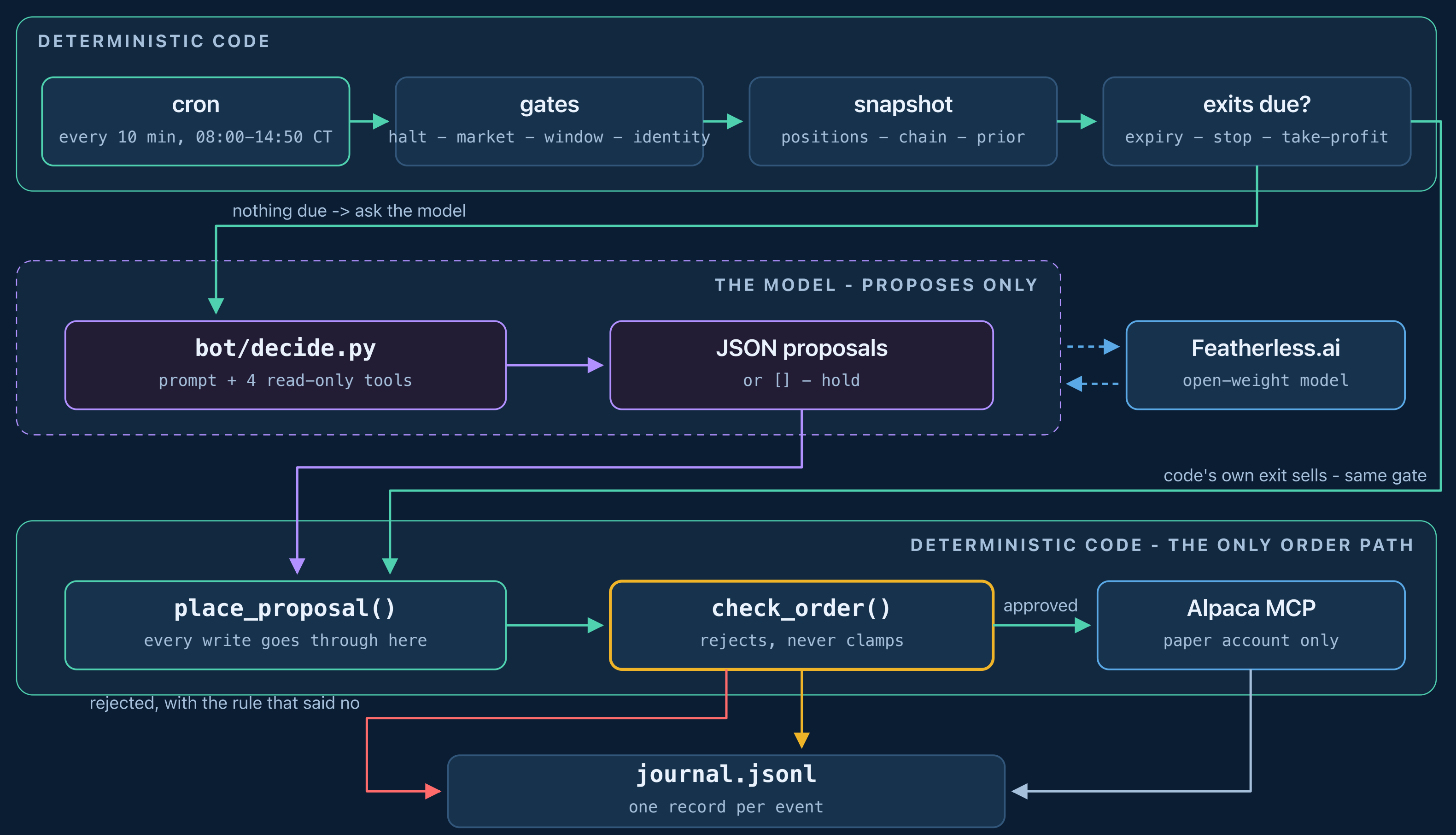
DEFINED RISK

Long premium only. The worst case is the premium paid, known before entry — which is what lets every guardrail be a flat number.

DIVISION OF LABOUR

The model forms a directional view from evidence. It never touches an order. Open model proposes; code disposes.

One cycle, every 10 minutes



Everything outside the dashed box is deterministic code - there is no path from the model to an order.

The model gets one box. Its proposals and code's own exit sells pass the same gate.

Greeks: Alpaca's, ours as the backstop

We spent three days sure the free feed carried none. It was wrong.

FROM ALPACA

IV and the full Greeks on **5,721 of 6,114** contracts in today's chain — one vendor surface, so a strike's put and call deltas agree.

BLACK-SCHOLES BACKSTOP

A bisection IV solve covers the ~6% too thin to price. Every contract carries `greeks_source`, so the model knows which numbers are rough.

What the model may choose from

One API page is 500 contracts. On SPY, that is three days.

	before	now
SPY chain	500 contracts, 1 page, to 3 DTE	2,912 contracts, 3 pages, to 45 DTE
NVDA menu	12 contracts, all one strike	at-the-money and ~0.40 delta, three expiries, both sides

Coverage is journaled each cycle, so a truncated chain is loud rather than silent.

A second opinion

```
SPY  prior close 7,712, median 7,688 (−0.31%); P(above) 0.293; volume 756  -> shown
QQQ  prior close 29,433, median 29,450 (+0.06%); P(above) 0.514; volume 45  ->
withheld
```

TWO CROWDS

Kalshi's index-close market and the option chain's own implied odds. A prior to weigh, not a signal to copy — no code acts on it.

KNOWING WHEN TO IGNORE IT

A barely-traded market quotes every bucket just as confidently. Volume and entropy gates withheld QQQ, **with the reason**.

…and then we grade it

A prior nobody scores is decoration. Brier score, nightly, against what the market actually did.

Tue Sep 1, judged account	Brier	
Kalshi index-close market	**0.004**	confident, and right
The option chain's own odds	**0.008**	agreed, less sharply
Priors the gate *withheld*	0.006	shadow-graded: the gate discarded nothing good

A coin flip scores 0.25. One day, and a directional one — the point is that the inputs are graded at all.

It investigates before it acts

```
research: get_bars({'symbol': 'SPY', 'timeframe': '15Min'})   -> 3334 chars
research: get_news({'symbols': 'NVDA', 'limit': 5})           -> 1770 chars
model output: [{"symbol":"NVDA260902P00220000","side":"buy","qty":10,
  "reason":"NVDA in a clear intraday downtrend (225 -> 218.6); the 220 put
            with 4 DTE, delta -0.58, aligns with the bearish momentum"}]
```

Four read-only tools, six calls at most, every one journaled. Nothing that places an order is reachable.

The leash

Per proposal	whitelist · ≤ \$5,000 per position · ≤ 4 positions · ≤ 10 contracts · 2–45 DTE on entries · entries 09:45–15:15 ET
Resting orders	an unfilled buy counts as held, and the next cycle cancels it
Before the model	close on expiry day · stop –40% · take-profit +60%
Per day	2% loss → flatten and halt · `HALT` file is a global kill switch

```
REJECTED buy 12 SPY260904C00775000: qty 12 exceeds max_contracts_per_order 10
```

Everything is journaled

`journal.jsonl`

`one per account, append-only`

every decision, order, rejection, exit, tool call and config hash

THE BROWSER

Live journal viewer

`bot.wpmccormick.pw`

- journal_viewer.py, :8300 on the LAN
- -> cloudflared tunnel
- -> Cloudflare Access (email OTP)
- server-sent events, live follow
- read-only: no controls at all

THE DASHBOARD

Home Assistant

`MQTT auto-discovery`

- journal.log() -> fire-and-forget
- equity, P&L, halt, last decision
- operator view: kill switch + knobs
- team view over Tailscale: read-only
- phone: problems only, never fills

THE INBOX

Hourly email

`mail_report.py`

- trades with the model's full reason
- CSVs attached, host local time
- silent when nothing traded
- one final send after the close

THE LOOP

eod_review.py

`the daily change`

- round trips from Alpaca's fills
- rejections grouped by rule
- Brier score on the priors
- critique by a model that did not trade

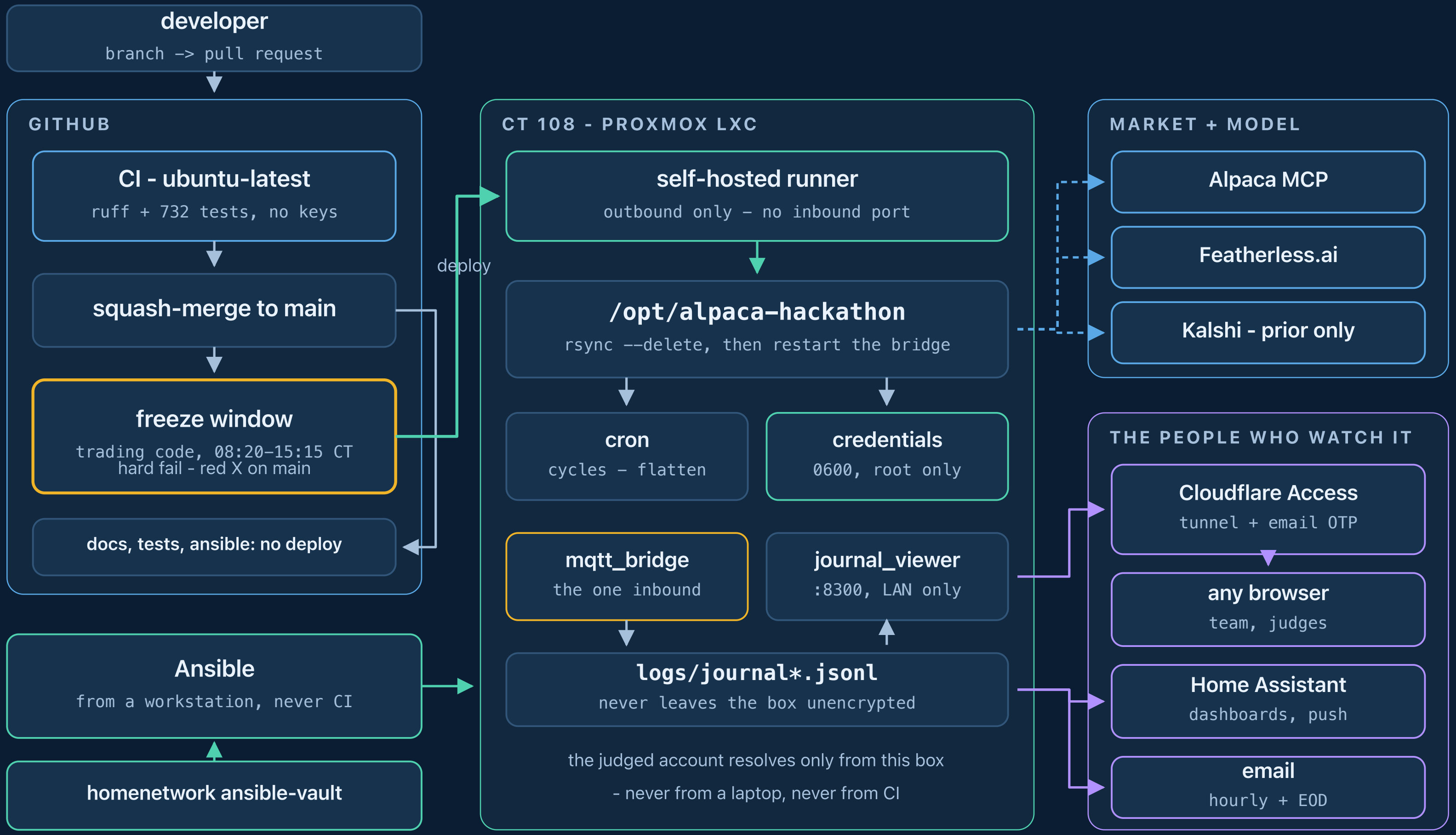
One file, four windows. Three are read-only views for humans; the fourth decides what changes tomorrow.

Watching it think, from anywhere

bot.wpmccormick.pw — the journal streamed to a browser, published through a Cloudflare tunnel. No VPN, nothing to install. It found four bugs in two days; none was a failing test.

#174	the feed cut the model's reasoning mid-word
#170	closing a +$1,340 winner, it named a *neighbouring strike*, three cycles running
#171	an unfilled limit sat invisible; the same thesis was bought again next door
#173	a ten-minute-old position proposed for exit — every number it cited had moved *in its favour*

Our own hardware, not a laptop



A merge deploys in about a minute. Nothing reaches the container inbound except the tunnel and the broker.

Built to be changed while it runs

735 tests, 2s	every one runs with **no API keys and no network**
Only Docker needed	tests, a full dry-run cycle, the docs site
Rehearse any time	`--dry-run` prints orders instead of sending; `--force` skips the market gates
No deploy needed	`override.py` — allowlisted knobs, hard caps stay git-only, all of it expires at the close

Dozens of those tests exist because something broke. Each incident ends as a test.

The daily loop

16:05 `eod_review` → round trips · rejections by rule · Brier → a second model's read → one change → override or config PR → deployed before the open

The critique comes from a model that did *not* trade the day. 132 PRs, 113 deploys, in seven days.

Real A/B, not a backtest

```
run_cycle.py --account official
run_cycle.py --account test  --config config-test.yaml
run_cycle.py --account mixed --config config-variants/mixed.yaml
```

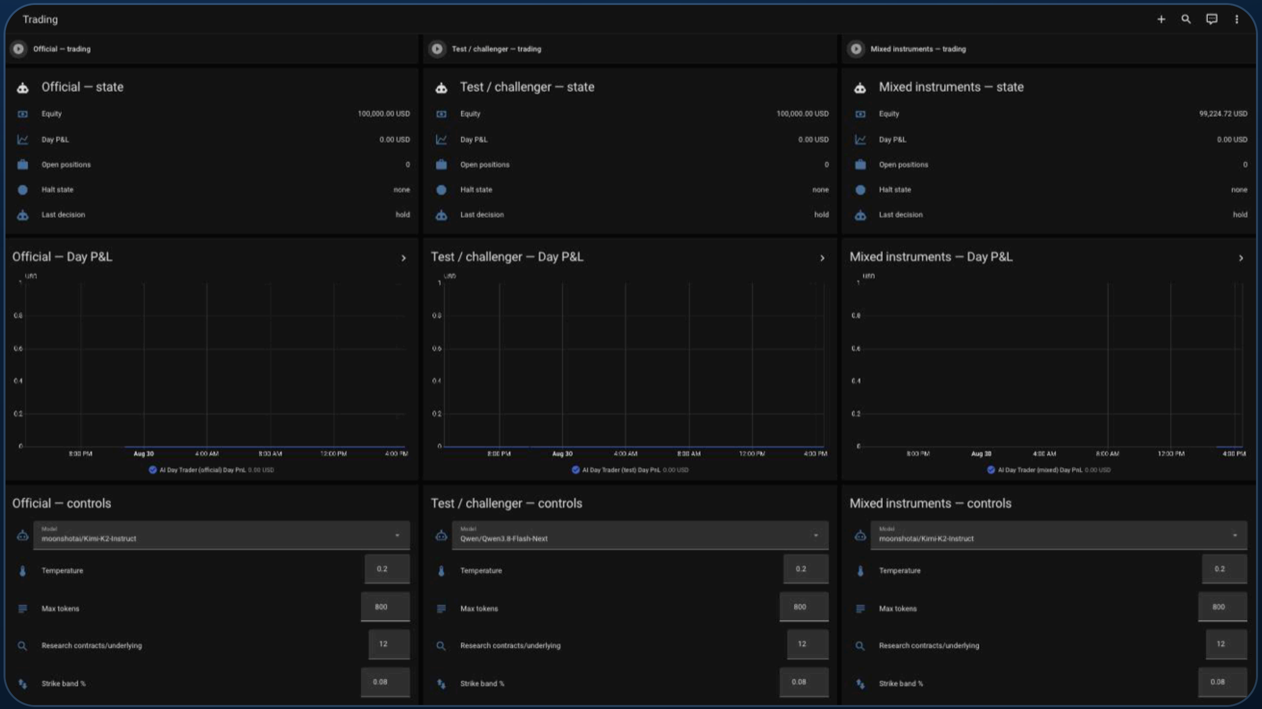
THREE LIVE ACCOUNTS

Same box, same ten-minute cadence, same market. Four days is no backtest, and the free feed has no historical options data.

ONE DELIBERATE DIFFERENCE EACH

`test` swaps the model and enables research and learning. `mixed` differs by exactly one key. The winner is promoted by pull request.

One feed, three audiences



OPERATOR

Kill switch and knobs.

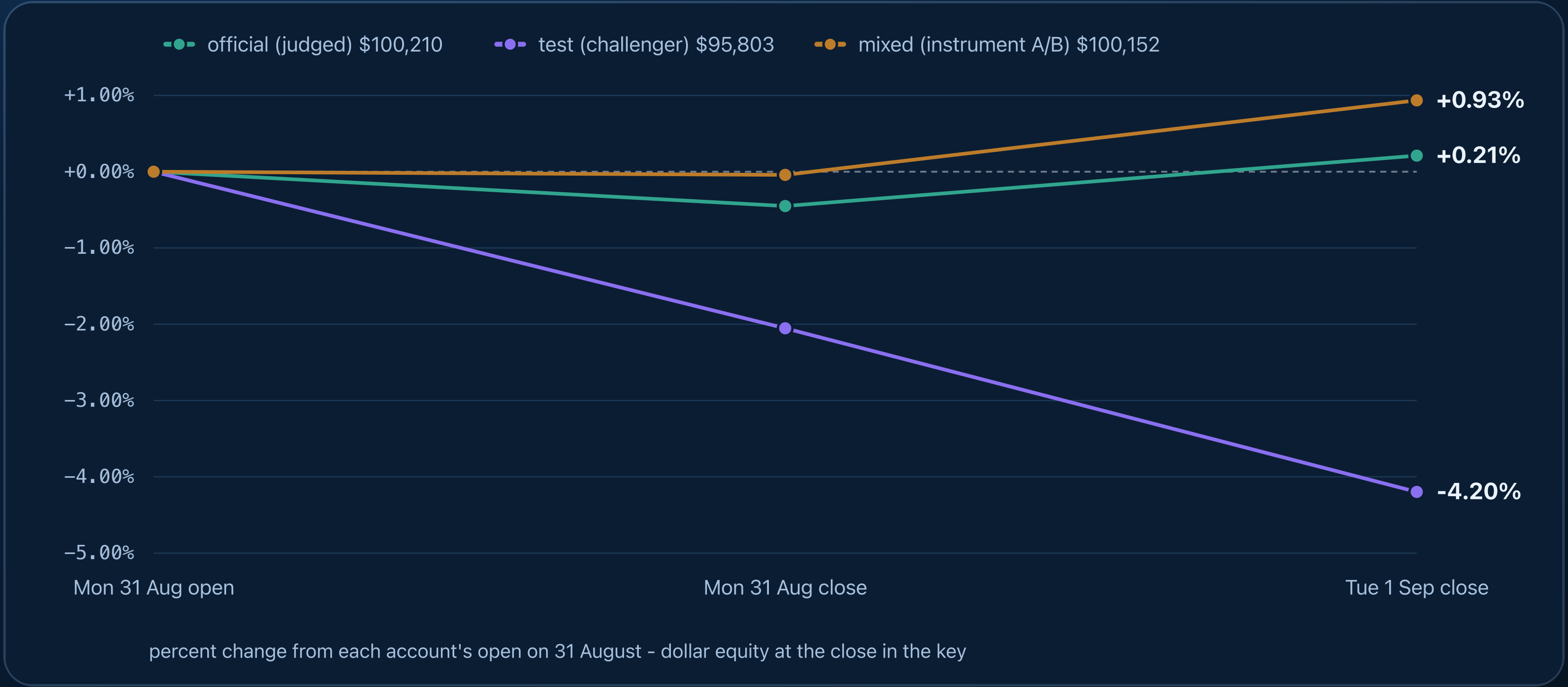
TEAM

A read-only dashboard — Home Assistant has no per-entity permissions, so the separation has to *be* the dashboard.

PHONE

Problems only. A fill sends nothing; *silence* does — no cycle for 25 minutes.

Results (fill Thu Sep 3)



Judged account Mon open → Thu close: $___ → $___ (__%) · __ round trips · what didn't work: __

What's next

PROMOTE WHAT THE CHALLENGERS PROVED

Research tools, the prediction-market prior, the learning loop.

MULTI-LEG SPREADS

Once the gates cover assignment risk.

github.com/bill-mccormick-dg/alpaca-hackathon · MIT
Thanks to Alpaca, lablab.ai and Featherless AI.